

Aditya Aryan

B.Stat student and researcher

✉ adityaharshit012345@gmail.com

📍 Kolkata, West Bengal, India

🌐 aditya-aryan-8049b0282

🔗 DudeDope

Profile

Bachelor of Statistics student at the Indian Statistical Institute, Kolkata, interested in mathematical statistics, probabilistic modelling, machine learning, and optimisation under uncertainty, with experience across theoretical research and applied modelling.

Education

Bach- Indian Statistical Institute, Statistics
elor of • Expected graduation: 2027.

Kolkata, West Bengal
July 2023 – 2027

**Statis-
tics (B.Stat),
First
Divi-
sion
with
Hon-
ours
(projected)**

CBSE Kendriya Vidyalaya No. 2, Jalahalli East, Physics, Chemistry, and Mathematics
Se- • Class XII: 95%.
nior • Class X: 98.8%.

Bengaluru, Karnataka
Apr 2014 – Mar 2023

**Secondary
and
Secondary
Edu-
cation**

Experience

University of Toronto, Student Researcher

Research with Prof. Xin Bing on expectation-maximisation for k-component Gaussian mixture-type models.

Remote
Apr 2026 – present
5 months

- Studying population and sample EM under nonconvex geometry, mixture imbalance, and high-dimensional noise.
- Analysing population maps, fixed points, and local contraction using Stein's lemma and Fisher/missing-information decompositions.
- Developing concentration-style bounds that connect population dynamics with finite-sample behaviour.

Ranial Systems, Data Science Intern

Applied machine learning, probabilistic modelling, and optimisation for energy-system monitoring, forecasting, and electricity-market decision support.

- Developing degradation and performance models, machine-learning and deep-learning methods for anomaly and alert prediction, and survival-modelling workflows.
- Applying vine-copula methods to model dependence between solar generation and weather variables and support probabilistic forecasting.
- Studying stochastic market-clearing-price (MCP) models and quantile-based bidding strategies for electricity-market decisions.
- Building Python pipelines for time-aware evaluation, backtesting, operational optimisation under battery constraints, and decision support.

Remote
May 2026 – present
4 months

Mercity AI, Machine Learning Research Intern

Compact multimodal modelling for medical image caption generation.

- Connected medSigLIP and Qwen-2.5-7B through a Q-Former, compressing 256 image tokens into 32 latent queries.
- Achieved an $8\times$ visual-token compression and approximately 70% memory reduction while preserving probabilistic calibration in the frozen-language-model setup.
- Trained the Q-Former with cross-entropy on MIMIC captions and explored SFT, PPO, DPO, and GRPO with the Hugging Face trl stack.
- Extended a DeepSeek-R1 GRPO workflow to multimodal inputs; recorded improved BLEU and lower perplexity in the reported experiments.

Bengaluru
Apr 2025 – Aug 2025
5 months

Projects

Stein's Paradox: Inadmissibility and Risk-Optimal Shrinkage

Supervised by Dr. Ayanendranath Basu, Indian Statistical Institute.

- Studied inadmissibility of the usual estimator for multivariate normal means when $d \geq 3$.
- Compared James–Stein and mean-centred shrinkage, heteroscedastic whitening, Normal–Normal empirical Bayes, Monte Carlo risk, and SURE.
- Recovered uniform James–Stein dominance and observed the lowest heteroscedastic simulation risk for the MMLE estimator.

Bivariate Copula Modelling of Extreme Air Pollution Events

Supervised by Dr. Shyamal Krishna De, Indian Statistical Institute.

- Modelled duration and severity of AQI episodes in Bengaluru from 2015–2024 using six copula families and nonparametric marginals.
- Estimated Kendall's $\tau = 0.881$ and Spearman's $\rho = 0.972$; the selected Gumbel model had $\theta = 6.183$ and upper-tail dependence 0.881.
- Obtained a model-derived estimate $P(S > 2000 \text{ given } D \geq 12) \approx 0.9998$ under the project's definitions.

Sequential Testing: Optimal Stopping under a Reward–Cost Trade-off

Supervised by Dr. Arnab Chakraborty, Indian Statistical Institute.

- Formulated a Bellman recursion over the state $S(x, y)$ and implemented memoised dynamic programming with backtracking and cutoffs.
- For $N = 20$ and $C = 5$, the specified instance produced an expected gain of 7.6875.

Variance-Based Audio Denoising via Amplitude Thresholding

Supervised by Dr. Arnab Chakraborty, Indian Statistical Institute.

- Estimated local variance over approximately 0.1-second windows and used a modal variance-bin cutoff for suppression and scaling.
- Implemented the workflow in R with time-domain and FFT-based diagnostics.

Elo-Based Football Betting

Supervised by Dr. Ayanendranath Basu, Indian Statistical Institute.

- Modelled English Premier League outcomes from 2014–2024 with Elo features and logistic regression.
- Recorded a Brier score of 0.2176 and log loss of 0.6249.
- A historical value-betting simulation returned 5,139 units of profit on a 10,000-unit test bankroll; this is a retrospective simulation, not a claim of future profitability.

Skills

Mathematical and statistical methods: Probability, Markov processes, High-dimensional statistics, Hypothesis testing, Linear modelling, Time series

Programming and data: Python, R, PyTorch, scikit-learn, statsmodels, pandas, NumPy, SciPy

Machine-learning tooling: Hugging Face trl, Accelerate, bitsandbytes, PEFT

Machine-learning topics: Transformers, Vision transformers, CNNs, Diffusion models, Deep reinforcement learning, Computer vision, Language modelling

Coursework

- Real Analysis; Probability Theory; Stochastic Processes; Statistical Methods; Linear Algebra; Differential Equations
- Programming and Data Structures; Molecular Biology; Psychology

Honors and Awards

- Indian Statistical Institute Entrance Examination (2023) — All India Rank 90
- Joint Entrance Examination (Main) (2023) — 98.9 percentile among approximately 1.5 million candidates
- BITS Admission Test (2023) — 331/390
- Kishore Vaigyanik Protsahan Yojana, SA stream (2022) — All India Rank 1,034
- Indian National Mathematical Olympiad (2022) — Merit list
- Indian Olympiad Qualifier in Mathematics (2022) — Karnataka top 20
- National Talent Search Examination Scholarship (2021) — Top 1,000 among approximately 2 million candidates
- National Talent Search Examination, Stage I (2021) — Rank 72

Academic Interests

Statistics and probabilistic modelling: Mathematical statistics, Statistical inference, Latent-variable models, Probabilistic modelling, High-dimensional statistics

Machine learning and decision methods: Machine learning and deep learning, Optimisation, Sequential decision-making, Forecasting and uncertainty quantification, Reliable evaluation of predictive models